AuraSpend

Native Android App & Standalone Database Architecture

Project Overview & Java Implementation Guide

- Transitioned from PC MySQL/Spring Boot dependency to 100% standalone Android app.
- Written in pure Java with XML layouts and native Android Material Components.
- Embedded SQLite database (auraspend.db) provides complete offline data independence.
- Identical feature set and visual parity with the luxury AuraSpend dark violet theme.
- Custom animated boot-up splash screen with pulsating halo and 3D squircle logo.

Executive Transformation

From Server-Dependent to 100% Standalone

Key Architectural Shifts

- OLD PROTOTYPE: Required running Spring Boot backend and MySQL 8 server on a host PC.
- LIMITATION: Android emulator or device failed with ’Can’t reach server’ if PC was offline.
- NEW NATIVE ARCHITECTURE: All accounts, expenses, and settings stored in device SQLite.
- ZERO EXTERNAL DEPENDENCIES: Zero cloud reliance, zero PC database, works fully offline.
- DATA PORTABILITY: Full JSON backup import/export compatible with web AuraSpend v2.

Database Architecture

Embedded SQLite Database (auraspend.db)

Relational Schema & Multi-User Isolation

- DATABASE HELPER: DatabaseHelper.java extends SQLiteOpenHelper with singleton pattern.
- TABLE 'users': Stores unique email, display name, currency, and SHA-256 hashed password.
- TABLE 'expenses': Stores user_email, amount, date (YYYY-MM-DD), category, and description.
- TABLE 'settings': 1-to-1 user settings storing preferred currency, budget, and theme mode.
- INDEXES: idx_expenses_user and idx_expenses_date guarantee sub-millisecond query performance.
- TRANSACTION SAFETY: importBackup uses beginTransaction() and setTransactionSuccessful().

Domain & Business Logic

Models, Currencies and Categories

Java Class Design

- CURRENCY (Currency.java): Supports 9 global currencies with dynamic symbols and formatting:
- - USD ($), INR (Rs.), EUR (E), GBP (L), AED (AED), CAD (C$), AUD (A$), JPY (Y), SGD (S$).
- CATEGORY (Category.java): 6 categorized spending buckets with emojis and labels:
- - Food (Burger), Transport (Car), Rent (House), Fun (Game), Utilities (Bulb), Others (Star).
- USER & EXPENSE (User.java, Expense.java): Clean POJOs for financial entity mapping.
- SESSION (SessionManager.java): Persistent SharedPreferences session store for auto-login.

UI Engine: Boot & Dashboard

Splash Choreography & Canvas Graphics

Interactive Screen Implementations

- BOOT ANIMATION (SplashActivity.java): Multi-stage startup choreography:
- - Ambient glow orbs fade in -> 3D logo springs with OvershootInterpolator bounce.
- - Radial halo breathing pulse animation -> Title & subtitle slide up -> Crossfade transition.
- CIRCULAR BUDGET RING (CircularProgressView.java): Custom Android Canvas 2D view:
- - Draws background ring track, progress arc, percentage text, and 'budget' label in center.
- DASHBOARD METRICS: Current month spending, remaining budget, daily average, top category.
- FILTER CHIPS: Interactive horizontally scrollable chips to filter category breakdown list.

UI Engine: History & Sheets

Searchable History & Slide-Up Sheets

Transaction Management Workflows

- HISTORY SCREEN: Real-time search by description or category using TextWatcher.
- DATE GROUPING: Automatically groups transactions into day buckets with weekday headers.
- ADD EXPENSE SHEET: Slide-up BottomSheetDialog with 3x2 interactive category selection grid.
- DATE PICKER: Native Android calendar DatePickerDialog integration defaults to current date.
- EDIT EXPENSE SHEET: Pre-fills values, allows in-place updating or deletion with confirmation alert.

Reports & Data Portability

On-Device PDF Engine & Backup System

Insights & Export Capabilities

- PERIOD TABS: Instant filtering for Weekly, Monthly, Annual, and All-time timeframes.
- PDF GENERATION: Uses Android’s native android.graphics.pdf.PdfDocument engine directly on device.
- SHARING INTEGRATION: Generates formatted report with table and shares via FileProvider intent.
- JSON BACKUP: Exports complete backup file (auraspend-backup.json) for cross-device migration.
- JSON RESTORE: Parses backup files, validates fields, and transactionally imports into SQLite.

Settings, Security & Verification

Theming, Protection and Build Quality

Implementation Quality & Testing

- APPEARANCE TOGGLE: Instant switching between Dark Obsidian and Light Lavender themes.
- LOCAL SECURITY: Data sandboxed in private app directory; SHA-256 one-way password hashing.
- SECURE SHARING: FileProvider grants temporary URI read permission without broad storage access.
- VERIFICATION: Built and tested with Gradle 8.9 and JDK 17 (BUILD SUCCESSFUL).
- UNIT TESTS: CurrencyAndCategoryTest passes 100% of test cases for calculations and hashing.